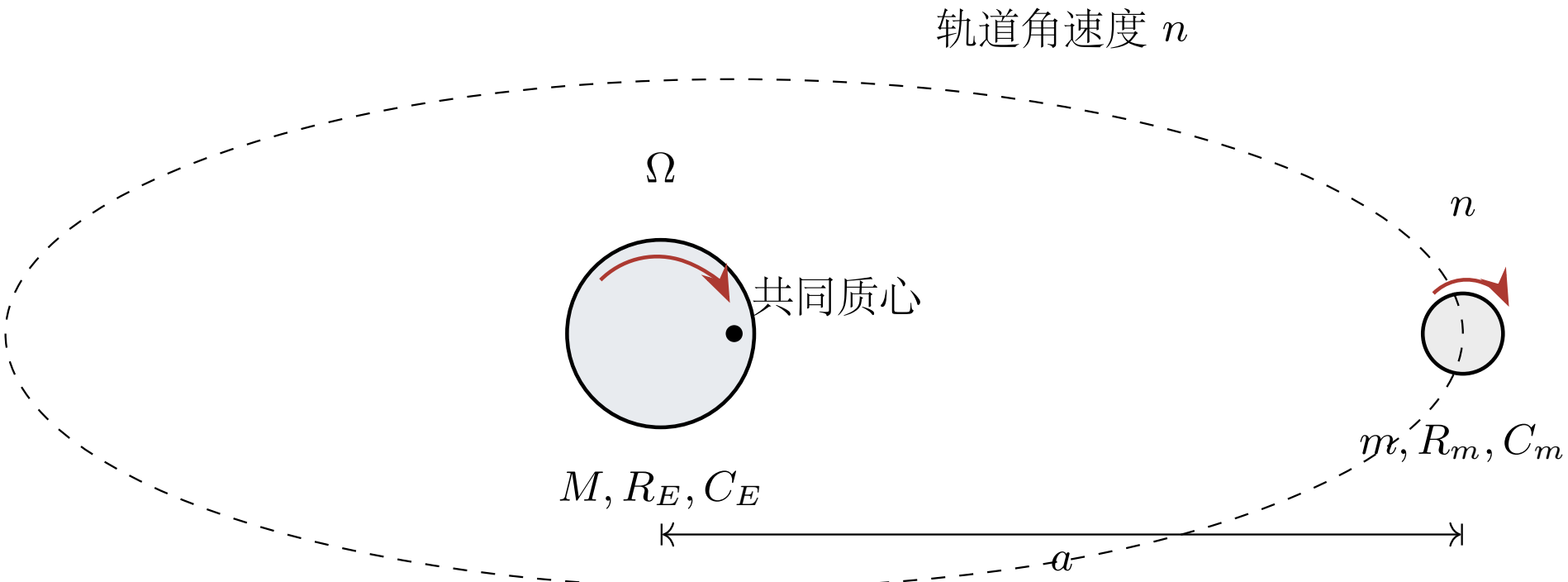

轨道角速度 n
Ω
n
共同质心
m, R_m, C_m
M, R_E, C_E
a

总角动量 L_0 守恒，机械能耗散

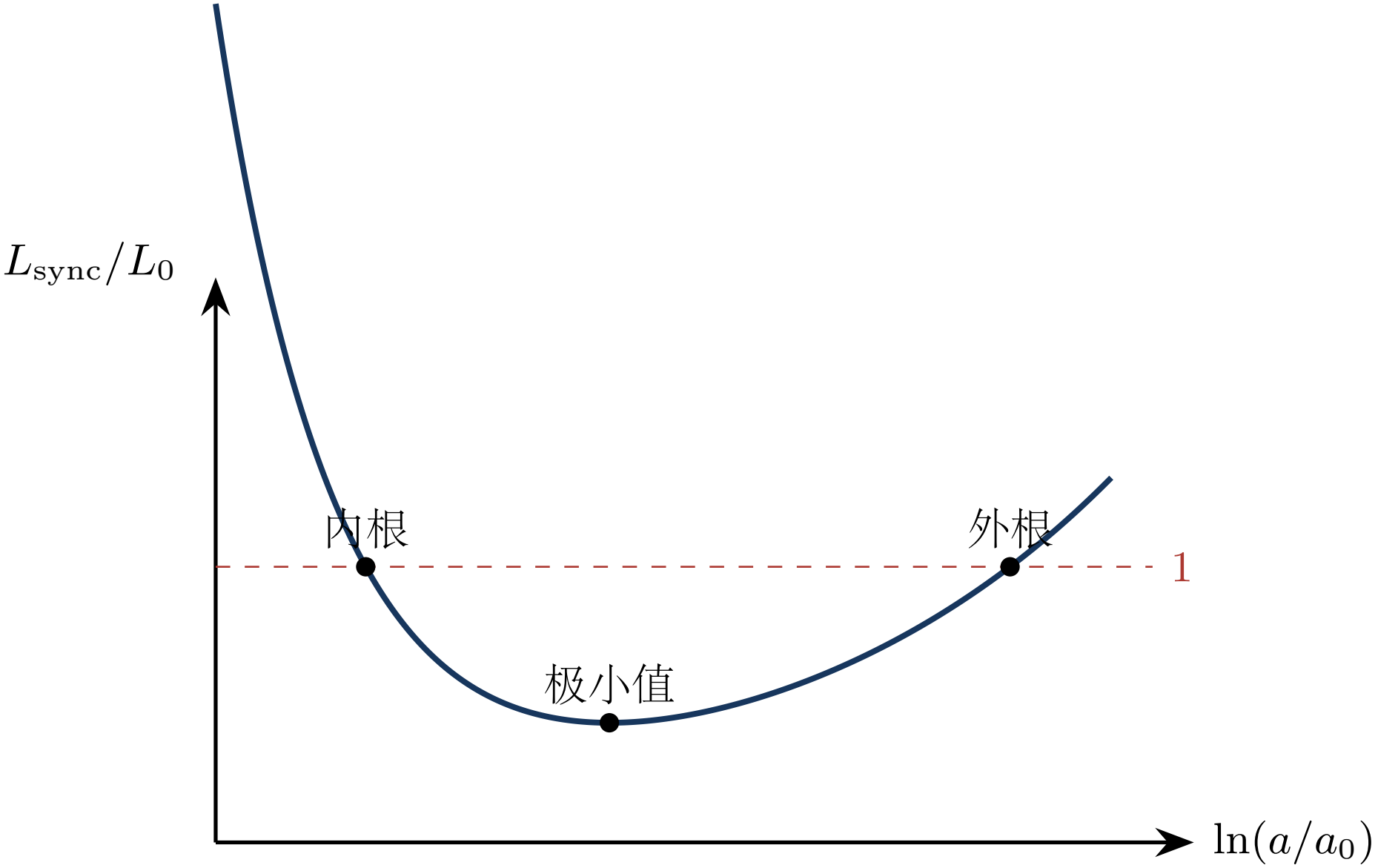
地球自转
$C_E\Omega$

潮汐转移

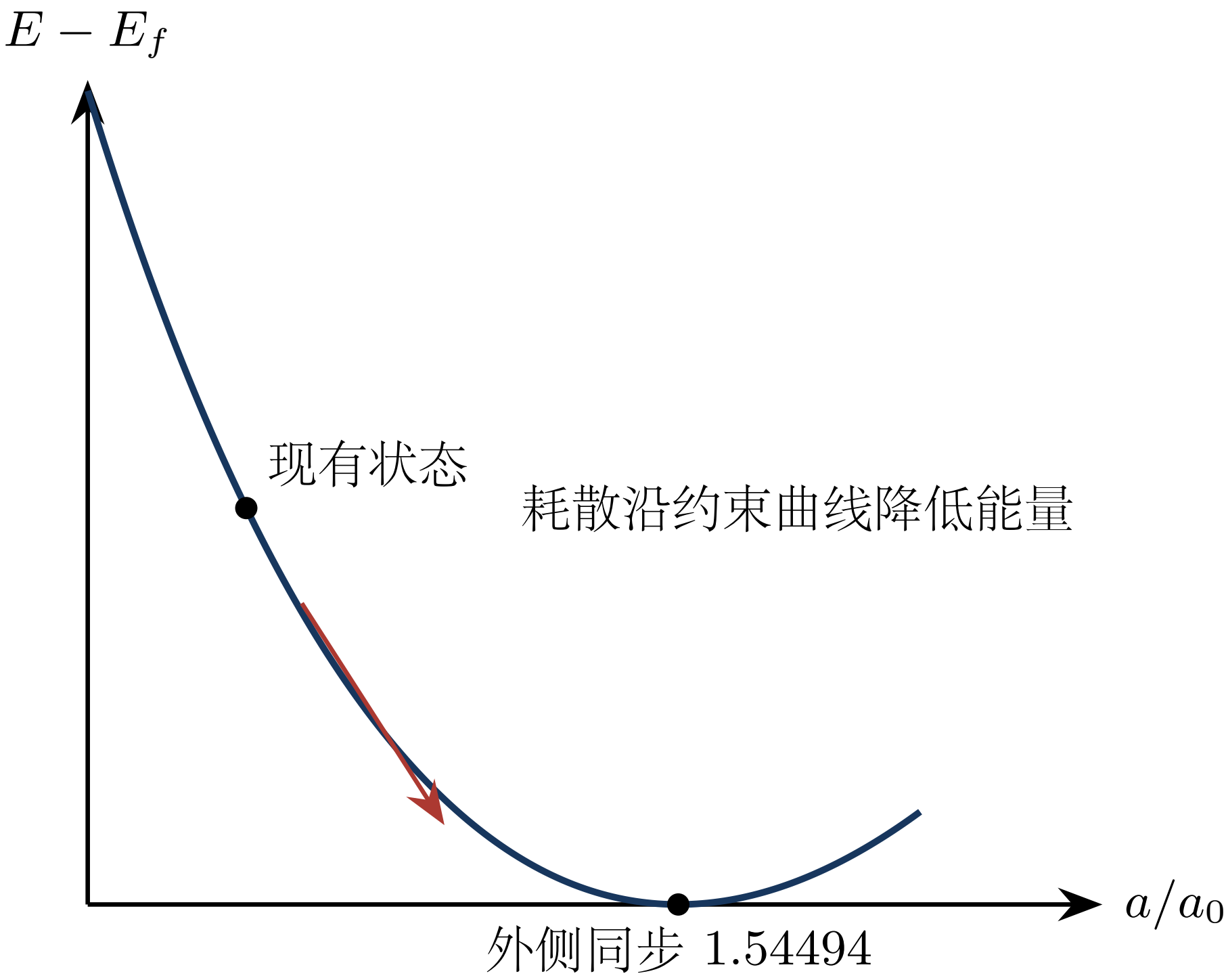
圆轨道
$\mu a^2 n \propto \sqrt{a}$

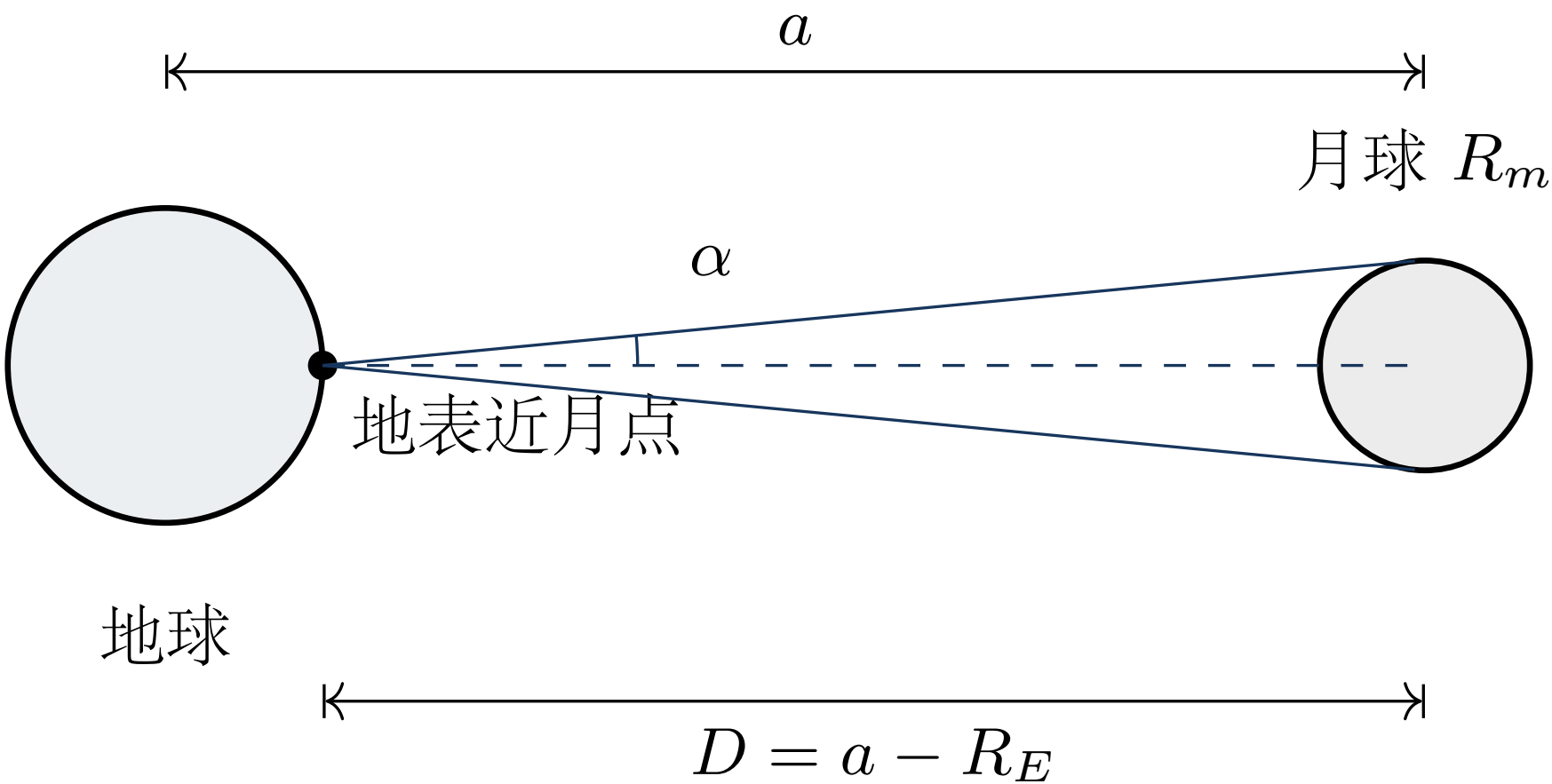
$a\uparrow, n\downarrow, L_{\mathrm{orb}}\uparrow$

月球同步自转
$C_m n$



横轴采用对数以同时显示两根


$E - E_f$
现有状态
耗散沿约束曲线降低能量
a/a_0
外侧同步 1.54494



$$\mathcal{O} = 2\pi(1 - \cos\alpha) \simeq \pi R_m^2 / D^2$$